

Graphs and characteristics of quadratic functions



- 1 Consider the equation $y = 25 - (x + 2)^2$.
 - a Does y have a minimum or maximum value?
 - b State the maximum value of y .

- 2 The maximum value of the function $y = -7x^2 - 140x + m$ is $y = 3$.
 - a Find the value of x at the point where the maximum value of the function occurs.
 - b Find the value of m .

- 3 Consider the function $f(x) = x^2 - 8x$.
 - a Does the function have any maxima or minima?
 - b At what value of x is $f(x)$ at its minimum?
 - c State the minimum value of the function.
 - d On which interval is the function increasing?
 - e On which interval is the function decreasing?

- 4 Consider the function $f(x) = x^2 + 4x + 7$.
 - a Sketch the graph of the function.
 - b Does the function have any maxima or minima?
 - c State the minimum value of the function.
 - d Over which interval is the function increasing?
 - e Over which interval is the function decreasing?

- 5 State whether the following descriptions could represent the graph of $f(x) = 9x^2(x + k) - x - k$, given that k is an integer.
 - a As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$, as $x \rightarrow \infty$, $f(x) \rightarrow \infty$, and the graph has $3x$ -intercepts.
 - b As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$, as $x \rightarrow \infty$, $f(x) \rightarrow \infty$, and the graph has $4x$ -intercepts.
 - c As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$, as $x \rightarrow \infty$, $f(x) \rightarrow \infty$, and the graph has $4x$ -intercepts.
 - d As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$, as $x \rightarrow \infty$, $f(x) \rightarrow \infty$, and the graph has $3x$ -intercepts.

- 6 Consider the general quadratic equation $y = ax^2 + bx + c$, $a \neq 0$:

- a** Is it correct to say that the vertex of the parabola will be located at $x = \frac{b}{2a}$?
- b** What should the x value of the vertex actually be?

7 Consider the function $y = x^2$

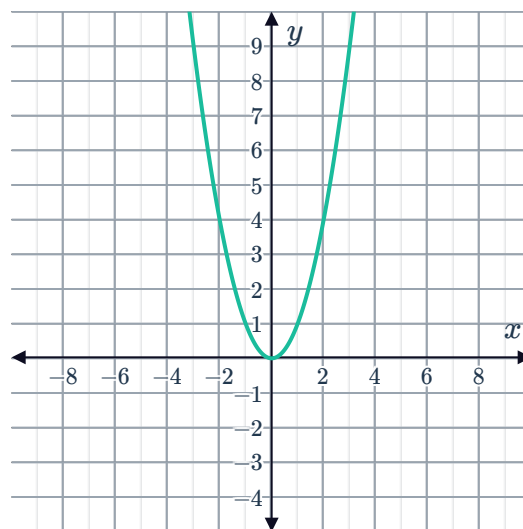
- a** Complete the following table of values.

x	-3	-2	-1	0	1	2	3
y							

- b** Plot the curve.
- c** Are the y values ever negative?
- d** Write down the equation of the axis of symmetry.
- e** State the minimum y value.
- f** For every y value greater than 0, how many corresponding x values are there?

8 Consider the graph of $y = x^2$.

- a** How do we shift the graph of $y = x^2$ to get the graph of $y = (x - 2)^2$?
- b** Hence plot $y = (x - 2)^2$ on the same graph as $y = x^2$.



9 Consider the equation $y = x^2 + 5$.

- a** Complete the set of solutions for the given equation.

x	-3	-2	-1		1	2	3
y				5			

- b** Plot the curve that results from the entire set of solutions for the equation being



10 Consider the function $y = (x - 6)^2$.

a Complete the table of values.

x	4	5	6	7	8
y					

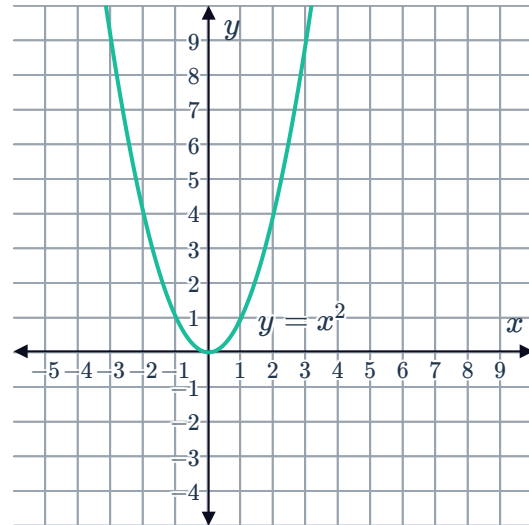
b State the minimum value of y .

c State the x value that corresponds to the minimum y value.

d Hence state the coordinates of the vertex of the parabola.

e How many units to the right has $y = x^2$ been translated?

f Translate the parabola $y = x^2$ to sketch $y = (x - 6)^2$.



11 Consider the function $y = -(x - 5)^2$.

a Complete the table of values.

x	3	4	5	6	7
y					

b State the maximum value of y .

c State the x value that corresponds to the maximum y value.

d Hence state the vertex of the parabola.

e Sketch the graph of $y = -(x - 5)^2$.

12 Consider the following equations:

i Find the x -value of the x -intercept.

ii Find the y -value of the y -intercept.

iii Determine the vertex.

iv Graph the equation.

a $y = (x - 2)^2$

b $y = (x - 5)^2 - 9$

13 Consider the function $y = x^2 - 6x + 5$.

- a Find the x -value of the x -intercepts of the function.
- b Sketch the graph of $y = x^2 - 6x + 5$.
- c Plot the points corresponding to $y = 0$ on the curve.

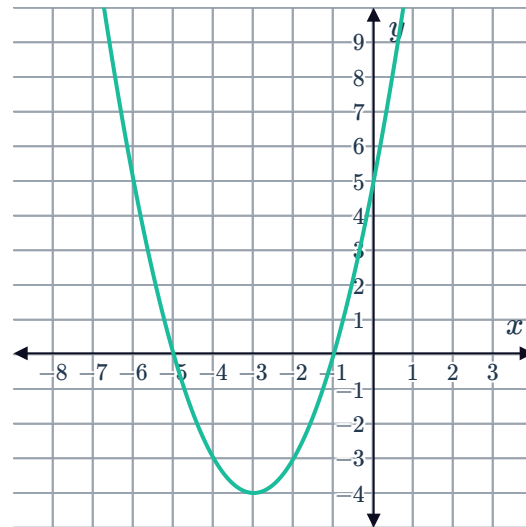


14 Consider the curve $y = x^2 + 2x - 2$.

- a State the equation of the axis of symmetry.
- b Now state the minimum value of y .
- c Using the minimum point on the curve, sketch the graph of the function.

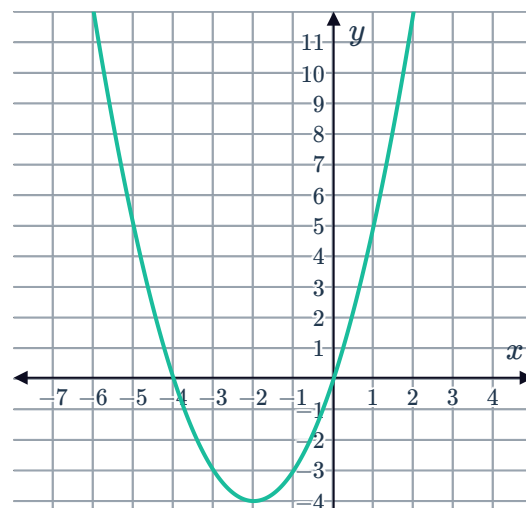
15 Consider the function $y = x^2 + 6x + 5$.

- a State the equation of the axis of symmetry.
- b State the minimum value of y .
- c Hence state the coordinates of the vertex of the curve.
- d Consider the graph of another quadratic function shown on the right. State the coordinates of its vertex.
- e State whether the following statements are correct about the two graphs.
 - i They have the same concavity.
 - ii They have the same x -intercepts.
 - iii They share the same turning point.



16 Consider the graph of the function $y = f(x)$:

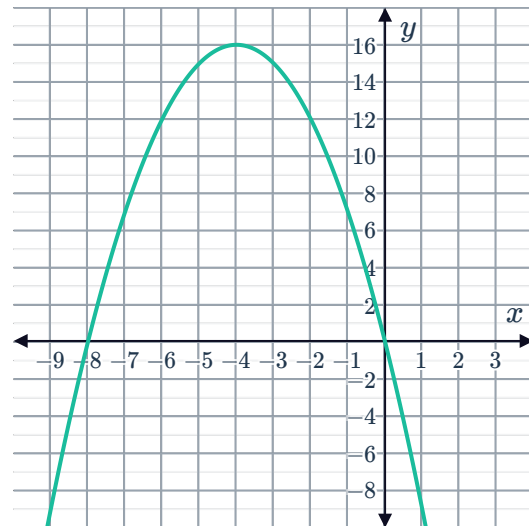
- a State the y -value of the absolute minimum of the graph.
- b Hence state the range of the function.
- c State the domain of this function.





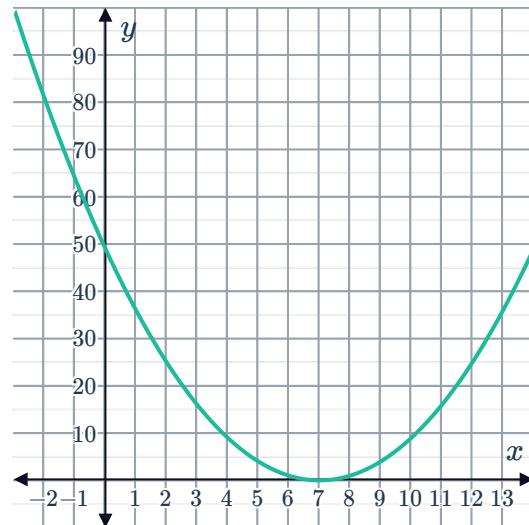
17 Consider the graph of the function $y = f(x)$:

- a State the absolute maximum of the graph.
- b Hence state the range of the function.
- c State the domain of this function.



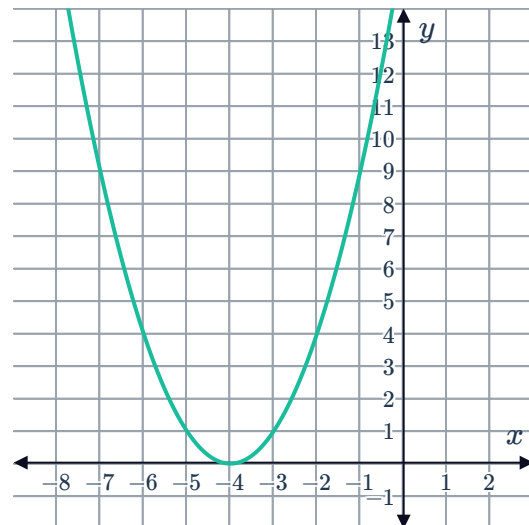
18 Consider the graph of the function $y = f(x)$:

- a State the absolute minimum of the graph.
- b Hence determine the range of the function.
- c Over what interval of the domain is the function increasing?



19 Consider the graph of the function $y = f(x)$ and answer the following questions.

- a State the absolute minimum of the graph.
- b Hence determine the range of the function.
- c Over what interval of the domain is the function decreasing?





20 Consider a quadratic function that passes through the points shown in the following table.

x	1	2	3	4	5	6	7
y	17	7	1	-1	1	7	17

- a Does this quadratic function have a maximum or minimum point?
- b State the maximum or the minimum value of the function.
- c State the range of the quadratic function.

21 Consider the table of coordinates that satisfy a particular quadratic function.

x	-9	-8	-7	-6	-5	-4	-3
y	-11	-6	-3	-2	-3	-6	-11

- a Does the function have a maximum or minimum point?
- b State the maximum or the minimum value of the function.
- c State the range of the function. Give your answer as an inequality.

22 A quadratic function includes the points shown in the table below.

x	5.5	6	6.5	7	7.5	8	8.5
y	15	5	-1	-3	-1	5	15

- a Does the function have a maximum or minimum point?
- b State the maximum or the minimum value of the function.
- c State the range of the function.

23 Consider a quadratic function that passes through the points shown in the following table.

x	-5.5	-5	-4.5	-4	-3.5	-3	-2.5
y	-12	-7	-4	-3	-4	-7	-12

- a Does the function have a maximum or minimum point?
- b State the maximum or the minimum value of the function.



24 Consider the equation $y = 2x^2$.

- a** Is every value of y positive?
- b** Consider the graph of $y = 2x^2$ shown on the right. State the axis of symmetry of the parabola.
- c** State the minimum value of y .

